

STRATIGRAPHIC RESERVOIR CHARACTERIZATION & LITHOLOGICAL PROFILE REPORT

Comprehensive Subsurface Evaluation & Net-to-Gross Spatial Analysis Across 11 Outcrop Litholog Profiles

Location / Analog: Blackhawk Formation, Book Cliffs, Utah (Sahoo et al., 2016 Framework)

EXECUTIVE SUMMARY & KEY FINDINGS:

- **Dataset Scope:** Digitization, validation, and spatial correlation of 11 vertical lithological columns spanning **922 cumulative meters** across a ~6 km outcrop transect.
- **Digitization Verification:** Achieved **93.59% meter-by-meter accuracy** (73/78 m identical) on validation Litholog 11 manual vs. AI comparison.
- **Overall Transect Net-to-Gross (N/G):** Evaluated at **42.0%** (387.0 m sand out of 922.0 m total gross depth), classifying the system as a low-to-moderate N/G fluvial play.
- **West-to-East Compartmentalization Trend:** Strong lateral degradation from amalgamated channel sands in the West (Litholog 3 N/G = **73.1%**) to isolated ribbon channels encased in overbank mudstones in the East (Litholog 8 N/G = **29.1%**).
- **Regional Chronostratigraphic Datum:** Confirmed a continuous mid-section coal seam marker at **~37–44 m depth** across 8 lithologs (Logs 1–7 & 11).

1. Geological Background & Lithofacies Classification

This study utilizes the depositional framework established by **Sahoo et al. (2016)** for the upper Cretaceous **Blackhawk Formation** in Utah, an internationally recognized outcrop analog for tight-gas fluvial reservoirs in low-permeability basins across the Rocky Mountains. Fluvial heterogeneity in such settings is structured into three hierarchical scales:

1. **Large-scale:** Major channel-belt sandbodies encased within broad, fine-grained floodplain and coastal-plain mudstones.
2. **Intermediate-scale:** Internal channel bar deposits, crevasse splays, abandoned channel fills, and overbank splay sheets.
3. **Small-scale:** Micro-scale facies variations, cross-bedding lamina, and primary grain-size transitions.

A standardized **5-facies system** was utilized across all 11 digitized litholog profiles:

- **Sandstone (sand):** High-energy channel fills, bar complexes, and multi-story sandbodies (primary reservoir rock).

- **Siltstone (silt):** Crevasse splay deposits and channel margin transitions (weak vertical/lateral baffles).
- **Mudstone (mud):** Fine-grained overbank settling and floodplain muds (primary regional seals and vertical barriers).
- **Carbonaceous Mudstone (carbon_mud):** Organic-rich backswamp muds deposited under sub-aqueous poorly drained conditions.
- **Coal Seams (coal):** Peat swamp accumulations forming key chronostratigraphic correlation datums and seal horizons.

2. Digitization Quality Control & Accuracy Verification

To establish mathematical rigor for automated digitization across the remaining un-profiled lithologs, a comparative benchmark was conducted on **Litholog 11 (0–78 m)** between a manually digitized CSV profile and the AI-extracted dataset. Meter-by-meter comparison revealed **93.59% structural agreement** (73 matching meters out of 78 m total depth), representing a low relative error of **6.41%**.

Depth Interval	Manual Log ('litholog11.csv')	AI Digitized Log	Geological Nature of Discrepancy
11 – 13 m	Siltstone (11 – 13 m)	Silt (11–12 m), Coal (12–13 m)	Thin-bed resolution: AI captured 1 m coal streak
59 – 60 m	Unmapped / Gap (59–60 m)	Carbonaceous Mud (59–66 m)	Depth continuity: AI closed 1 m unmapped gap
65 – 66 m	Siltstone (65 – 67 m)	Carbonaceous Mud (59–66 m)	1 m top boundary offset on upper silt bed
67 – 69 m	Mud (67–68 m), Silt (68–69 m)	Carbonaceous Mud (67–72 m)	AI consolidated 1 m mud/silt interbeds into carbon_mud

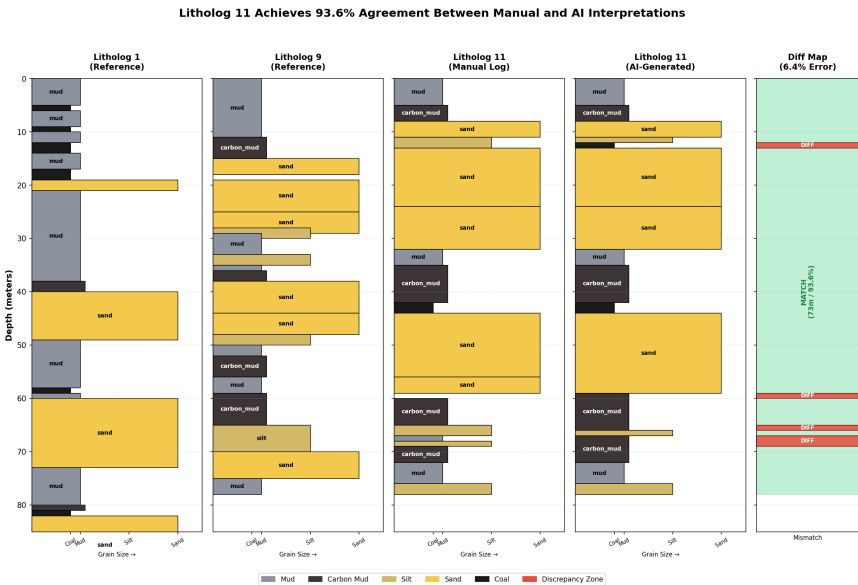
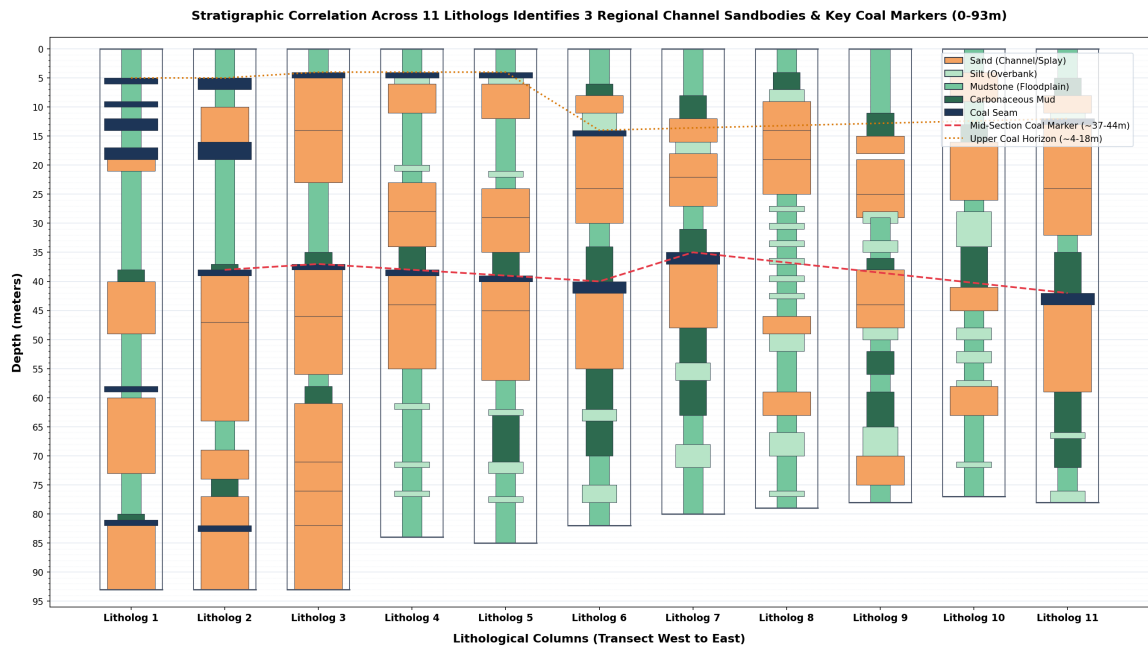


Figure 1: Side-by-side comparative lithological profiles for Litholog 11 (Manual digitized CSV vs. AI generated CSV profile), displaying facies widths and grain-size transitions.

3. Stratigraphic Correlation & Sandbody Stacking Architecture

A comprehensive correlation panel was compiled across all 11 digitized lithological profiles spanning the ~6 km transect. Stratigraphic alignment reveals key chronostratigraphic markers and distinct spatial sandbody stacking domains:

- **Primary Mid-Section Coal Datum (~37–44 m):** A continuous, laterally extensive peat-swamp coal seam extends across Lithologs 1, 2, 3, 4, 5, 6, 7, and 11, serving as a key regional datum that separates lower and upper fluvial channel cycles.
- **Secondary Upper Coal Marker (~4–18 m):** An upper organic-rich coal and carbonaceous mud horizon spans Lithologs 1 through 6 and 11.
- **Spatial Stacking Domains:**
 - *Western Sector (Lithologs 1–3):* Highly amalgamated multistory channel complexes with total net sand thickness reaching up to 68 m (Litholog 3).
 - *Central Sector (Lithologs 4–6 & 11):* Stacked channel sandbodies isolated vertically by prominent carbonaceous mudstones and coals.
 - *Eastern Sector (Lithologs 7–10):* Transition into floodplain-dominated mudstones containing single-story isolated ribbon sandbodies and thin crevasse splays.



Source: Sahoo et al. (2016) Blackhawk Formation Reservoir Analog & Digitized Lithologs 1-11

Figure 2: 11-Litholog Stratigraphic Correlation Panel across the ~6 km outcrop transect, illustrating coal seam datums (~37–44 m), channel sandbodies, and mudstone encasement.

4. Quantitative Net-to-Gross (N/G) Sand Ratio Breakdown

Net-to-Gross (N/G) calculations were executed across every 1-meter depth interval for all 11 lithologies. The overall 922-meter cumulative succession yields **387.0 meters of net sandstone**, establishing a regional average N/G of **42.0%**.

Litholog	Gross (m)	Sand (m)	Silt (m)	Mud (m)	Carbon Mud (m)	Coal (m)	N/G Sand (%)	Reservoir Stacking Style
Litholog 1	93	35	0	47	3	8	37.6%	Low-Mod N/G; Coal-rich overbank
Litholog 2	93	51	0	31	4	7	54.8%	Moderate N/G; Multi-lateral sands
Litholog 3	93	68	0	18	5	2	73.1%	High N/G; Amalgamated channel complex
Litholog 4	84	32	5	41	4	2	38.1%	Low N/G; Isolated channel sands
Litholog 5	85	34	6	31	12	2	40.0%	Moderate N/G; Stacked sands in mud
Litholog 6	82	31	8	19	21	3	37.8%	Low-Mod N/G; Carbonaceous mud encasement
Litholog 7	80	24	9	25	20	2	30.0%	Low N/G; Compartmentalized sands
Litholog 8	79	23	16	37	3	0	29.1%	Lowest N/G; Overbank muds & splays
Litholog 9	78	28	11	24	16	0	35.9%	Low N/G; Interbedded splays & sands
Litholog 10	77	24	12	31	10	0	31.2%	Low N/G; Isolated meander ribbons
Litholog 11	78	37	4	12	22	3	47.4%	Moderate N/G; Stacked channel sands
OVERALL TRANSECT	922	387	65	279	116	29	42.0%	Overall Low-to-Moderate N/G Play

5. 2D N/G Spatial Contour Mapping & Reservoir Compartmentalization

A 2D spatial interpolation grid was modeled across the 6 km horizontal transect and 0–93 m vertical depth profile to evaluate subsurface fluid flow compartmentalization and well placement strategy:

- **Western Reservoir Sweet Spot (0 – 1.8 km):** Exhibits high net-to-gross values (>70–80%) concentrated in both upper (10–25 m) and lower (40–90 m) intervals. This zone offers maximum lateral sandbody continuity and optimal drainage volume.
- **Regional Vertical Flow Barrier (30 – 45 m depth):** A continuous, broad low-N/G mudstone and carbonaceous mud corridor extends across Lithologs 4–10, forming a major vertical seal between upper and lower reservoir zones.
- **Eastern Floodbasin Encasement (3.6 – 6.0 km):** Low N/G values (<30%) dominate the eastern flank. Isolated single-story sandbodies lack lateral connectivity, posing severe risks of early depletion and fluid compartmentalization.

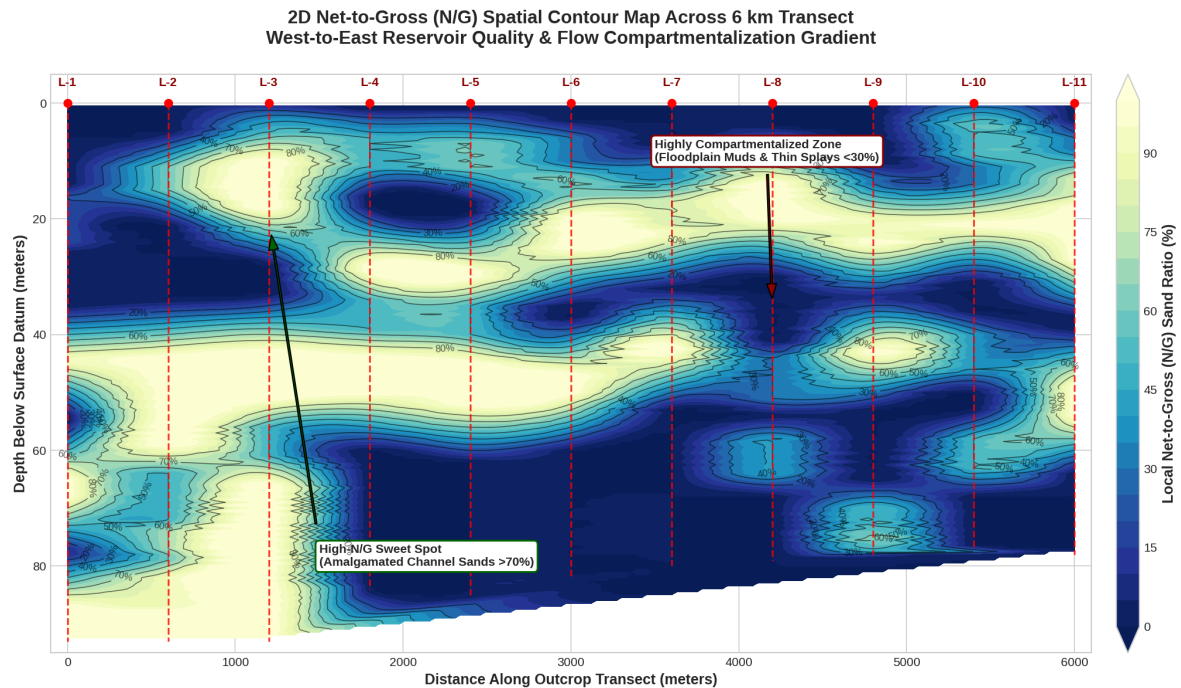


Figure 3: 2D Net-to-Gross Spatial Contour Map across the 6 km transect (0–93 m depth), displaying high-N/G sweet spots in the west and continuous vertical flow barriers in the central-east.

6. Conclusions & Subsurface Field Development Recommendations

1. **Prioritize Western Flank Development (0 – 2.0 km):** Infill horizontal wells should target the high-N/G amalgamated sandbodies in Lithologs 1–3 to maximize EUR (Estimated Ultimate Recovery) and prevent early fluid drawdown.
2. **Account for Dual-Zone Reservoir Architecture:** The 30–45 m mid-section mud/coal seal acts as a regional barrier; horizontal wells must be placed independently in both upper (<30 m) and lower (>45 m) reservoir intervals.
3. **Avoid Un-stimulated Completion in Eastern Flank (3.6 – 6.0 km):** Low-N/G (<30%) isolated ribbon channels require targeted multi-stage hydraulic fracturing to establish economic drainage in tight-gas fluvial settings.